

Opcodes, mnemonics and cycle counts generated live from genucode.py (the microcode source of truth) -- cannot drift from the hardware.

Operands: #imm immediate | addr 16-bit absolute | (Pn) ptr indirect | (Pn)+ post-increment | (no operand) implied. **By=**bytes, **Cy=**cycles (incl. fetch). **Flags (F):** C carry (active-high; ADD carry-out / SUB,CMP no-borrow A>=B), Z zero, N negative (bit7), V overflow, *- = none. Signed branches test N^V / Z. **Memory (rev E):** \$0000-1FFF ROM | \$2000-FEFF RAM | \$FF00-FFFF I/O. P0=PC, P3=stack (empty-descending).

JZ/JNZ/JC are aliases of BZ/BNZ/BCP.

Mnemonic						By						Cy						FI						Description																							
System																								Stack																							
\$90	NOP	1	2	-	No operation.	\$70	PHA	1	2	-	Push A onto P3 stack.	\$71	PLA	1	3	ZN	Pop A from P3 stack.																														
\$91	HLT	1	2	-	Halt clock; resume only by reset.	\$72	CLC	1	2	C	C←0.	16-bit memory (rev D)												\$74	PHW addr	3	9	-	Push 16-bit word at addr (h then lo: lies little-endian at P3+1).																		
\$72	CLC	1	2	C	C←0.	\$73	SEC	1	2	C	C←1.	\$75	PLW addr	3	10	-	Pop 16-bit word into addr (lo then hi).	\$76	LPW1 addr	3	9	-	P1←16-bit word at addr.																								
Interrupts (rev C)																								\$77	LPW2 addr	3	9	-	P2←16-bit word at addr.	\$78	MOVW dst, src	5	13	-	16-bit mem→mem: word at src→dst.	\$79	LPW3 addr	3	9	-	P3←16-bit word at addr (restore a saved SP).						
\$92	EI	1	2	-	Enable maskable interrupts (IE←1).	\$80	PHW (P1+d)	2	10	CZN	Push the word at P1+d (h then lo). A!; flags from the address add.	\$81	PHW (P2+d)	2	10	CZN	Push the word at P2+d. A!; flags from the address add.	\$82	PHW (P3+d)	2	10	CZN	Push the word at P3+d, 0 measured before the push (a localarg onto the stack). A!; flags from the address add.																								
\$93	DI	1	2	-	Disable maskable interrupts (IE←0).	\$83	PHW (P3+d)	2	10	CZN	Push the word at P3+d, 0 measured before the push (a localarg onto the stack). A!; flags from the address add.	Tier A: C-compiler ISA (2026-09, pure microcode; A! = clobbers A)												\$38	LDP1 #imm16	3	5	-	P1←imm16 (3 bytes, was the LPL1LPH1 pair).																		
\$94	RTI	1	8	-	Return from interrupt: pop flags then PC; re-enable IE.	\$84	PHW (P3+d)	2	10	CZN	Push the word at P3+d, 0 measured before the push (a localarg onto the stack). A!; flags from the address add.	\$39	LPD2 #imm16	3	5	-	P2←imm16.	\$3A	LPD3 #imm16	3	5	-	P3←imm16.																								
\$98	IRQ	1	10	-	SW interrupt: push PC+flags, vector to \$0808.	\$85	PHW (P3+d)	2	10	CZN	Push the word at P3+d, 0 measured before the push (a localarg onto the stack). A!; flags from the address add.	\$3C	ADDP3 #imm	2	6	CZN	P3→P3+imm8 (free a frame). A!; flags from the low byte.	\$30	SUBP3 #imm	2	6	CZN	P3→P3+imm8 (allocate a frame). A!; flags from the low byte.																								
Load/store																								\$30	SUBP3 #imm	2	6	CZN	P3→P3+imm8 (allocate a frame). A!; flags from the low byte.	\$99	LDW addr, (P1+J)	14	CZN	word at addr→word at P1+d. A!	\$91	LDW addr, (P2+J)	14	CZN	word at addr→word at P2+d. A!	\$92	LDW addr, (P3+J)	14	CZN	word at addr→word at P3+d (local→memory word). A!			
\$10	LDA #imm	2	2	ZN	A←immediate.	\$99	LDW addr, (P1+J)	14	CZN	word at addr→word at P1+d. A!	\$91	LDW addr, (P2+J)	14	CZN	word at addr→word at P2+d. A!	\$92	LDW addr, (P3+J)	14	CZN	word at addr→word at P3+d (local→memory word). A!	\$94	STW (P1+d), ad#J	14	CZN	word at P1+d→word at addr. A!																						
\$11	LDB #imm	2	2	ZN	B←immediate.	\$92	LDW addr, (P3+J)	14	CZN	word at addr→word at P3+d (local→memory word). A!	\$94	STW (P1+d), ad#J	14	CZN	word at P1+d→word at addr. A!	\$95	STW (P2+d), ad#J	14	CZN	word at P2+d→word at addr. A!	\$96	STW (P3+d), ad#J	14	CZN	word at P3+d→word at addr (memory word→local). A!																						
\$12	LDA addr	3	6	ZN	A←byte at addr (absolute).	\$94	STW (P1+d), ad#J	14	CZN	word at P1+d→word at addr. A!	\$95	STW (P2+d), ad#J	14	CZN	word at P2+d→word at addr. A!	\$96	STW (P3+d), ad#J	14	CZN	word at P3+d→word at addr (memory word→local). A!	\$98	LDW addr, #imm8	8	-	word at addr←imm8 zero-extended (4 bytes).																						
\$13	LDB addr	3	6	ZN	B←byte at addr (absolute).	\$95	STW (P2+d), ad#J	14	CZN	word at P2+d→word at addr. A!	\$96	STW (P3+d), ad#J	14	CZN	word at P3+d→word at addr (memory word→local). A!	\$98	LDW addr, #imm8	8	-	word at addr←imm8 zero-extended (4 bytes).	\$99	LDW addr, #imm16	9	-	word at addr←imm16 (5 bytes).																						
\$14	STA addr	3	6	-	byte at addr←A (absolute).	\$96	STW (P3+d), ad#J	14	CZN	word at P3+d→word at addr (memory word→local). A!	\$98	LDW addr, #imm8	8	-	word at addr←imm8 zero-extended (4 bytes).	\$99	LDW addr, #imm16	9	-	word at addr←imm16 (5 bytes).	\$9A	ADDW dst, src	5	15	CZNV	word a←a+b, 16-bit; C←carry out. A!; Z from the high byte only.																					
\$15	LDA (P1)+	1	2	ZN	A←[P1], P1++.	\$98	LDW addr, #imm8	8	-	word at addr←imm8 zero-extended (4 bytes).	\$99	LDW addr, #imm16	9	-	word at addr←imm16 (5 bytes).	\$9A	ADDW dst, src	5	15	CZNV	word a←a+b, 16-bit; C←carry out. A!; Z from the high byte only.	\$9B	SUBW dst, src	5	15	CZNV	word a←a-b, 16-bit; C←1 no borrow (a←b unsigned). A!; Z high byte only.																				
\$16	LDA (P2)+	1	2	ZN	A←[P2], P2++.	\$99	LDW addr, #imm16	9	-	word at addr←imm16 (5 bytes).	\$9A	ADDW dst, src	5	15	CZNV	word a←a+b, 16-bit; C←carry out. A!; Z from the high byte only.	\$9B	SUBW dst, src	5	15	CZNV	word a←a-b, 16-bit; C←1 no borrow (a←b unsigned). A!; Z high byte only.	\$9C	CPMW dst, src	5	15	CZNV	flags from a←b (16-bit), memory unchanged: C←unsigned a>b, BLT/BGE = signed. A!																			
\$17	LDA (P3)+	1	2	ZN	A←[P3], P3++.	\$9A	ADDW dst, src	5	15	CZNV	word a←a+b, 16-bit; C←carry out. A!; Z from the high byte only.	\$9B	SUBW dst, src	5	15	CZNV	word a←a-b, 16-bit; C←1 no borrow (a←b unsigned). A!; Z high byte only.	\$9C	CPMW dst, src	5	15	CZNV	flags from a←b (16-bit), memory unchanged: C←unsigned a>b, BLT/BGE = signed. A!	\$9E	INCM addr	3	9	CZN	word at addr←+1. A!; flags from the low byte.																		
\$19	STA (P1)+	1	2	-	[P1]←A, P1++.	\$9B	SUBW dst, src	5	15	CZNV	word a←a-b, 16-bit; C←1 no borrow (a←b unsigned). A!; Z high byte only.	\$9C	CPMW dst, src	5	15	CZNV	flags from a←b (16-bit), memory unchanged: C←unsigned a>b, BLT/BGE = signed. A!	\$9E	INCM addr	3	9	CZN	word at addr←+1. A!; flags from the low byte.	\$9F	DECM addr	3	9	CZN	word at addr←-1. A!; flags from the low byte.																		
\$1A	STA (P2)+	1	2	-	[P2]←A, P2++.	\$9C	CPMW dst, src	5	15	CZNV	flags from a←b (16-bit), memory unchanged: C←unsigned a>b, BLT/BGE = signed. A!	\$9E	INCM addr	3	9	CZN	word at addr←+1. A!; flags from the low byte.	\$9F	DECM addr	3	9	CZN	word at addr←-1. A!; flags from the low byte.	\$A0	ADDW addr, #imm8	15	CZNV	word a←a+imm8 (zero-extd, 16-bit; C←carry out; Z of the FULL word. A!)																			
\$1B	STA (P3)+	1	2	-	[P3]←A, P3++.	\$9E	INCM addr	3	9	CZN	word at addr←+1. A!; flags from the low byte.	\$9F	DECM addr	3	9	CZN	word at addr←-1. A!; flags from the low byte.	\$A0	ADDW addr, #imm8	15	CZNV	word a←a+imm8 (zero-extd, 16-bit; C←carry out; Z of the FULL word. A!)	\$A1	SUBW addr, #imm8	15	CZNV	word a←a-imm8; C←1 no borrow; Z full word. A!																				
\$1D	STA (P1)	1	2	-	[P1]←A.	\$9F	DECM addr	3	9	CZN	word at addr←-1. A!; flags from the low byte.	\$A0	ADDW addr, #imm8	15	CZNV	word a←a+imm8 (zero-extd, 16-bit; C←carry out; Z of the FULL word. A!)	\$A1	SUBW addr, #imm8	15	CZNV	word a←a-imm8; C←1 no borrow; Z full word. A!	\$A2	CPMW addr, #imm8	15	CZNV	flags from a←imm8 (16-bit), memory unchanged; Z full word (a←imm). A!																					
\$1E	STA (P2)	1	2	-	[P2]←A.	\$A0	ADDW addr, #imm8	15	CZNV	word a←a+imm8 (zero-extd, 16-bit; C←carry out; Z of the FULL word. A!)	\$A1	SUBW addr, #imm8	15	CZNV	word a←a-imm8; C←1 no borrow; Z full word. A!	\$A2	CPMW addr, #imm8	15	CZNV	flags from a←imm8 (16-bit), memory unchanged; Z full word (a←imm). A!	\$A4	LEAW addr, (P1+J)	14	CZN	word at addr→P1+d (the address of a frame local). A!																						
\$1F	STA (P3)	1	2	-	[P3]←A.	\$A1	SUBW addr, #imm8	15	CZNV	word a←a-imm8; C←1 no borrow; Z full word. A!	\$A2	CPMW addr, #imm8	15	CZNV	flags from a←imm8 (16-bit), memory unchanged; Z full word (a←imm). A!	\$A4	LEAW addr, (P1+J)	14	CZN	word at addr→P1+d (the address of a frame local). A!	\$A5	LEAW addr, (P2+J)	14	CZN	word at addr→P2+d. A!																						
\$51	LDA (P1)	1	2	ZN	A←[P1] (P1 kept).	\$A2	CPMW addr, #imm8	15	CZNV	flags from a←imm8 (16-bit), memory unchanged; Z full word (a←imm). A!	\$A4	LEAW addr, (P1+J)	14	CZN	word at addr→P1+d (the address of a frame local). A!	\$A5	LEAW addr, (P2+J)	14	CZN	word at addr→P2+d. A!	\$A6	LEAW addr, (P3+J)	14	CZN	word at addr→P3+d (address of a stack local). A!																						
\$52	LDA (P2)	1	2	ZN	A←[P2] (P2 kept).	\$A4	LEAW addr, (P1+J)	14	CZN	word at addr→P1+d (the address of a frame local). A!	\$A5	LEAW addr, (P2+J)	14	CZN	word at addr→P2+d. A!	\$A6	LEAW addr, (P3+J)	14	CZN	word at addr→P3+d (address of a stack local). A!	\$A1	ADDW addr, #imm16	15	CZNV	word a←a+imm16, C←carry out; Z full word. A!																						
\$53	LDA (P3)	1	2	ZN	A←[P3] (P3 kept).	\$A5	LEAW addr, (P2+J)	14	CZN	word at addr→P2+d. A!	\$A6	LEAW addr, (P3+J)	14	CZN	word at addr→P3+d (address of a stack local). A!	\$A1	ADDW addr, #imm16	15	CZNV	word a←a+imm16, C←carry out; Z full word. A!	\$B2	SUBW addr, #imm16	15	CZNV	word a←a-imm16; C←1 no borrow; Z full word. A!																						
\$88	LDA (P1+d)	2	7	CZN	A←byte at P1+d (d unsigned 0..255). C from the address add. Z/N from A.	\$A6	LEAW addr, (P3+J)	14	CZN	word at addr→P3+d (address of a stack local). A!	\$A1	ADDW addr, #imm16	15	CZNV	word a←a+imm16, C←carry out; Z full word. A!	\$B2	SUBW addr, #imm16	15	CZNV	word a←a-imm16; C←1 no borrow; Z full word. A!	\$B3	CPMW addr, #imm16	15	CZNV	flags from a←imm16, memory unchanged: C←a←imm unsigned, Z←a←imm, BLT/BGE signed. A!																						
\$89	LDA (P2+d)	2	7	CZN	A←byte at P2+d.	\$B2	SUBW addr, #imm16	15	CZNV	word a←a-imm16; C←1 no borrow; Z full word. A!	\$B3	CPMW addr, #imm16	15	CZNV	flags from a←imm16, memory unchanged: C←a←imm unsigned, Z←a←imm, BLT/BGE signed. A!	\$B4	ANDW dst, src	5	15	ZN	word a←a AND b; Z high byte only. A!																										
\$8A	LDA (P3+d)	2	7	CZN	A←byte at P3+d (a stack local).	\$B3	CPMW addr, #imm16	15	CZNV	flags from a←imm16, memory unchanged: C←a←imm unsigned, Z←a←imm, BLT/BGE signed. A!	\$B4	ANDW dst, src	5	15	ZN	word a←a AND b; Z high byte only. A!	\$B5	ORW dst, src	5	15	ZN	word a←a OR b; Z high byte only. A!																									
\$8C	STA (P1+d)	2	9	CZN	byte at P1+d←A. A kept, flags clobbered by the address add.	\$B4	ANDW dst, src	5	15	ZN	word a←a AND b; Z high byte only. A!	\$B5	ORW dst, src	5	15	ZN	word a←a OR b; Z high byte only. A!	\$B6	XORW dst, src	5	15	ZN	word a←a XOR b; Z high byte only. A!																								
\$8D	STA (P2+d)	2	9	CZN	byte at P2+d←A.	\$B5	ORW dst, src	5	15	ZN	word a←a OR b; Z high byte only. A!	\$B6	XORW dst, src	5	15	ZN	word a←a XOR b; Z high byte only. A!	\$B7	ANDW addr, #imm8	15	ZN	word a←a AND imm8 (high byte cleared); Z full word. A!																									
\$8E	STA (P3+d)	2	9	CZN	byte at P3+d←A (a stack local).	\$B6	XORW dst, src	5	15	ZN	word a←a XOR b; Z high byte only. A!	\$B7	ANDW addr, #imm8	15	ZN	word a←a AND imm8 (high byte cleared); Z full word. A!	\$B8	ORW addr, #imm8	15	ZN	word a←a OR imm8 (high byte kept); Z full word. A!																										
ALU (A←>A)																								\$B8	ORW addr, #imm8	15	ZN	word a←a OR imm8 (high byte kept); Z full word. A!	\$B9	XORW addr, #imm8	15	ZN	word a←a XOR imm8 (high byte kept); Z full word. A!	\$BA	ANDW addr, #imm16	15	ZN	word a←a AND imm16; Z full word. A!									
\$20	ADD	1	2	CZN	A←A+B.	\$B9	XORW addr, #imm8	15	ZN	word a←a XOR imm8 (high byte kept); Z full word. A!	\$BA	ANDW addr, #imm16	15	ZN	word a←a AND imm16; Z full word. A!	\$BB	ORW addr, #imm16	15	ZN	word a←a OR imm16; Z full word. A!	\$BC	XORW addr, #imm16	15	ZN	word a←a XOR imm16 (0xFFFF = bitwise NOT); Z full word. A!																						
\$21	SUB	1	2	CZN	A←A-B.	\$BA	ANDW addr, #imm16	15	ZN	word a←a AND imm16; Z full word. A!	\$BB	ORW addr, #imm16	15	ZN	word a←a OR imm16; Z full word. A!	\$BC	XORW addr, #imm16	15	ZN	word a←a XOR imm16 (0xFFFF = bitwise NOT); Z full word. A!																											
\$22	AND	1	2	CZN	A←A AND B.	\$BB	ORW addr, #imm16	15	ZN	word a←a OR imm16; Z full word. A!	Control flow														\$40	JMP addr	3	4	-	P0(PC)←addr.																	
\$23	OR	1	2	CZN	A←A OR B.	Control flow														\$41	JSR (P1)	1	9	-	Push return addr, P0←P1.																						
\$24	XOR	1	2	CZN	A←A XOR B.	\$40	JMP addr	3	4	-	P0(PC)←addr.	\$41	JSR (P1)	1	9	-	Push return addr, P0←P1.	\$42	RTS	1	6	-	Pop return addr from P3 into P0.																								
\$25	CMF	1	2	CZN	Flags from A-B; A unchanged.	\$41	JSR (P1)	1	9	-	Push return addr, P0←P1.	\$42	RTS	1	6	-	Pop return addr from P3 into P0.	\$43	JSR addr	3	10	-	Push return addr, P0←addr.																								
\$26	INC	1	2	CZN	A←A+1 (B unused).	\$42	RTS	1	6	-	Pop return addr from P3 into P0.	\$43	JSR addr	3	10	-	Push return addr, P0←addr.	\$48	BZ addr	3	4	-	Branch if Z=1. (JZ alias.)																								
\$27	DEC	1	2	CZN	A←A-1 (B unused).	\$43	JSR addr	3	10	-	Push return addr, P0←addr.	\$48	BZ addr	3	4	-	Branch if Z=1. (JZ alias.)	\$49	BNZ addr	3	4	-	Branch if Z=0. (JNZ alias.)																								
\$28	SHL	1	2	CZN	A←A<<1, 0→bit0, out→C.	\$48	BZ addr	3	4	-	Branch if Z=1. (JZ alias.)	\$49	BNZ addr	3	4	-	Branch if Z=0. (JNZ alias.)	\$4A	BCP addr	3	4	-	Branch if C=1 / A>=B unsigned. (JC alias.)																								
\$29	SHR	1	2	CZN	A←A>>1, 0→bit0, out→C.	\$49	BNZ addr	3	4	-	Branch if Z=0. (JNZ alias.)	\$4A	BCP addr	3	4	-	Branch if C=1 / A>=B unsigned. (JC alias.)	\$4C	JNC addr	3	4	-	Branch if C=0 / A<B unsigned.																								
\$2A	ROL	1	2	CZN	Rotate A left through carry.	\$4A	BCP addr	3	4	-	Branch if C=1 / A>=B unsigned. (JC alias.)	\$4C	JNC addr	3	4	-	Branch if C=0 / A<B unsigned.																														
\$2B	ROR	1	2	CZN	Rotate A right through carry.	\$4C	JNC addr	3	4	-	Branch if C=0 / A<B unsigned.													\$A8	JMP re18	2	9	CZNV	P0→P0+re18 (2-byte jump). A! flags; assembler relax / JMP.R. Always-taken jumps are cheaper absolute (JMP.A, 3 steps).																		
ALU with T (rev C; 2nd operand-T, B preserved)																								\$A8	JMP re18	2	9	CZNV	P0→P0+re18 (2-byte jump). A! flags; assembler relax / JMP.R. Always-taken jumps are cheaper absolute (JMP.A, 3 steps).	\$A9	BNZ re18	2	9	CZNV	Branch re18 if Z=1. Taken: A! flags! (JZ.R alias.)												
\$80	ADDT	1	2	CZN	A←A+T (B preserved).	\$A9	BNZ re18	2	9	CZNV	Branch re18 if Z=1. Taken: A! flags! (JZ.R alias.)	\$A9	BNZ re18	2	9	CZNV	Branch re18 if Z=1. Taken: A! flags! (JZ.R alias.)	\$AA	BNZ re18	2	9	CZNV	Branch re18 if Z=0. Taken: A! flags! (JNZ.R alias.)																								
\$81	SUBT	1	2	CZN	A←A-T (B preserved).	\$AA	BNZ re18	2	9	CZNV	Branch re18 if Z=0. Taken: A! flags! (JNZ.R alias.)	\$AA	BNZ re18	2	9	CZNV	Branch re18 if Z=0. Taken: A! flags! (JNZ.R alias.)	\$AB	BCP re18	2	9	CZNV	Branch re18 if C=1. Taken: A! flags! (JC.R alias.)																								
\$82	ANDT	1	2	CZN	A←A AND T (B preserved).	\$AB	BCP re18	2	9	CZNV	Branch re18 if C=1. Taken: A! flags! (JC.R alias.)	\$AB	BCP re18	2	9	CZNV	Branch re18 if C=1. Taken: A! flags! (JC.R alias.)	\$AC	JNC re18	2	9	CZNV	Branch re18 if C=0. Taken: A! flags!.																								
\$83	ORT	1	2	CZN	A←A OR T (B preserved).	\$AC	JNC re18	2	9	CZNV	Branch re18 if C=0. Taken: A! flags!.	Signed branches (rev C; after CMP)												\$44	BLT addr	3	4	-	Branch if signed A<B (N^V=1). After CMP.																		
\$84	XORT	1	2	CZN	A←A XOR T (B preserved).	Signed branches (rev C; after CMP)												\$44	BLT addr	3	4	-	Branch if signed A<B (N^V=1). After CMP.	\$45	BGE addr	3	4	-	Branch if signed A>=B (N^V=0). After CMP.																		
\$85	CMPT	1	2	CZN	Flags from A-T; A,B unchanged.	\$44	BLT addr	3	4	-	Branch if signed A<B (N^V=1). After CMP.	\$45	BGE addr	3	4	-	Branch if signed A>=B (N^V=0). After CMP.	\$46	BLE addr	3	4	-	Branch if signed A<=B (N^V=0). After CMP.																								
\$86	LOD #imm	2	2	-	T←immediate.	\$45	BGE addr	3	4	-	Branch if signed A>=B (N^V=0). After CMP.	\$46	BLE addr	3	4	-	Branch if signed A<=B (N^V=0). After CMP.	\$47	BGT addr	3	4	-	Branch if signed A>B. After CMP.																								
\$87	LOD addr	3	6	-	T←byte at addr (absolute).	\$46	BLE addr	3	4	-	Branch if signed A<=B (N^V=0). After CMP.	\$47	BGT addr	3	4	-	Branch if signed A>B. After CMP.	\$4D	BLT re18	2	9	CZNV	Branch re18 if signed A<B (N^V). Taken: A! flags!.																								
Pointer registers																								\$4D	BLT re18	2	9	CZNV	Branch re18 if signed A<B (N^V). Taken: A! flags!.	\$4E	BGE re18	2	9	CZNV	Branch re18 if signed A>=B. Taken: A! flags!.												
\$31	LPL1 #imm	2	3	-	P1 low byte←imm.	\$4E	BGE re18	2	9	CZNV	Branch re																																				